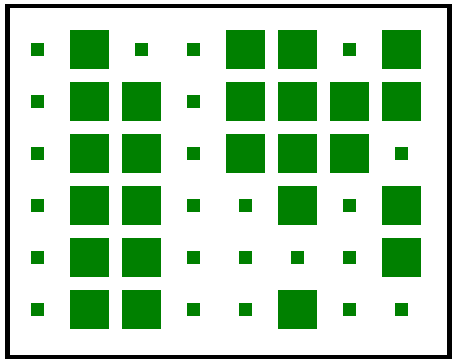
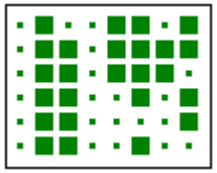




Lab 1
C Introduction

CS 330
Programming Languages and Compilers

Set Up Git Project for Team Work



- Open a terminal and change directory into your home directory:

```
cd ~
```

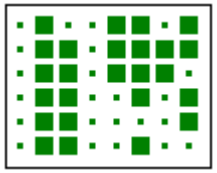
- Clone the team lab work project:

```
git clone TEAM_NAME@bits.monead.com:cs330-TeamLabWork
```

  - TEAM_NAME is one of: **alpha, ana, omega**

- You will be using this git project to submit all your team lab work for this class
- You will not clone it again, unless you want to make it available on another computer

LLM Usage for Lab 1

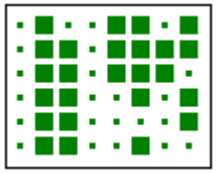


- Here is the policy: **You may change your mind after starting an exercise, except, you may not go from any LLM mode to the "I've got this!" mode because you've seen hints from the LLM**
  - In the lab README.md file, list the mode(s) used and, if you change modes, note why
- The requirement for using the LLM for at least ten exercises during the semester, with at least 4 being the "You do it!" mode, still apply
- For this lab, at least **three of the team and three of the individual exercises** must be done with the "I've got this!" mode (e.g., no LLM)

Lab Assignment

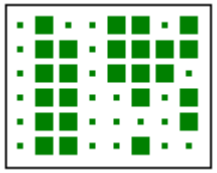
Team and Individual

Lab Assignment Naming



- For the problems on the following pages, (team and individual) each answer should be in its own file named `lab1prob-#.c`
  - Where # is the problem identifier
    - For instance, for team problem 1 the file will be named **`lab1prob-t-1.c`**
- For problems that require a text-answer (not a program), simply place the text answer within the file – it won't compile which is fine
- You submit your work via **`git`** as described at the end of the assignment

Team Lab Assignment



t-1)Explain the purpose of the characters \a, \b, \n, \t, and \\

t-2)Add a column header to the temperature conversion program (any version)

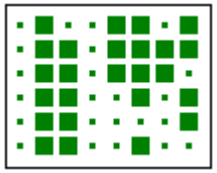
t-3)Create a program to convert Celsius to Fahrenheit (with a column header like problem t-2)

t-4)Write a program to print a histogram of digit counts (use digitcounter.c as a starting point)

t-5)Write a program which demonstrates that function parameters are passed by value (e.g. proves they are copies)

- Team lab 1 work goes in the **`cs330-TeamLabWork/lab1-C-Teamwork`** folder
- Add role assignments and LLM choices for exercises to the `README.md` file

Individual Lab Assignment



i-1)Leave out parts of the hello world program and collect some different error messages you receive when compiling

i-2)Write a program to prove that relational operators result in values 0 (for false) and 1 (for true)

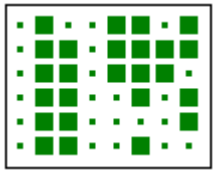
i-3)Write a program to print the value of `EOF`

i-4)Create a second version of the `power` function which uses recursion instead of iteration, printing results for bases 0-9, squared and cubed

- This work is done on your own
- Individual lab 1 work goes in the **`cs330-IndividualLabWork/lab1-C-Introduction`** folder
- Add LLM choices for exercises to the `README.md` file

Submitting the Team Lab Work

Submitting the Team Lab (1 of 2)



- The student who hosted the latest editing session, using the terminal, in the **`cs330-TeamLabWork`** directory, use the following commands:

  `git add .`

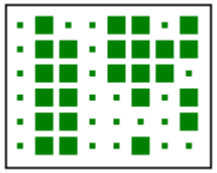
  - Note the trailing dot (preceded by a space) is needed

  `git commit -m "Updated lab work"`

  `git push`

  - The first time you push you may be asked to set the default branch. Enter the command as given in the message.

Submitting the Team Lab (2 of 2)



- After the code has been pushed (previous slide), the students who did not host the latest editing session should obtain the updated code, using the terminal, in the project directory, use the following command:

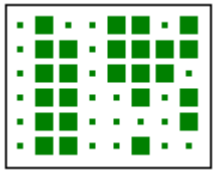
```
git pull
```

- When the team has completed the lab, the team's understanding leader for this lab creates and pushes an annotated tag to indicate the work is completed (the assignment isn't submitted until this is done):

```
git tag -a Lab1Submission1 -m "Lab completed"
git push --follow-tags
```

- Team members should receive an email within an hour verifying the lab was received for grading

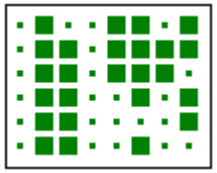
Errors When Using Git



- If an error occurs please capture a screenshot of the error and send that to the instructor along with a overview of what you were trying to do
  - Generally the errors are related to not being in the correct directory when using the Git commands

Submitting Individual Lab Work

Submitting the Individual Lab (1 of 2)



- Using the terminal, in the **`cs330-IndividualLabWork`** directory, use the following commands:

```
git add .
```

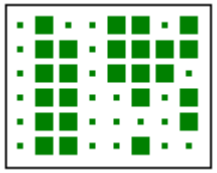
  - Note the trailing dot (preceded by a space) is needed

```
git commit -m "Updated lab work"
```

```
git push
```

  - The first time you push you may be asked to set the default branch. Enter the command as given in the message.

Submitting the Individual Lab (2 of 2)

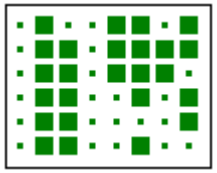


- When you have completed the lab, including following all steps on the previous page after your last changes are made, create and push an annotated tag to indicate the assignment is completed
- Assignment isn't submitted until this is done:

```
git tag -a Lab1Submission1 -m "Lab completed"
git push --follow-tags
```

- You should receive an email within an hour verifying the lab was received for grading

Errors When Using Git



- If an error occurs please capture a screenshot of the error and send that to the instructor along with a overview of what you were trying to do
  - Often the errors are related to not being in the correct directory when using the Git commands